Here is a comprehensive, in-depth breakdown of **Unit-3** and **Unit-4** tailored for 2-mark examination questions, providing clear, detailed, and technically precise answers for each item.

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UNIT-3: BASIC AND SPECIAL PURPOSE PROGRAMMABLE PERIPHERALS AND THEIR INTERFACING WITH 8086

1. What is semiconductor memory interfacing?

Semiconductor memory interfacing involves connecting memory ICs (RAM/ROM) to the 8086 address, data, and control buses. It requires:

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- **Address Decoding:** Selects the specific memory chip when its designated memory range is addressed.
- **Control Signaling:** Generates appropriate read ($\overline{\mathrm{MEMR}}$) or write ($\overline{\mathrm{MEMW}}$) signals to transfer data correctly.

2. Why is refreshing necessary in Dynamic RAM (DRAM)?

Dynamic RAM stores binary data as electrical charges in MOS transistor-capacitor cells. Because these capacitors slowly discharge due to leakage currents, the stored data degrades over time. Periodic refreshing (recharging every few milliseconds) reads and rewrites the stored charge to prevent data loss.

3. State the primary function of the 8255 Programmable Peripheral Interface (PPI).

The 8255 PPI acts as a general-purpose, software-configurable parallel I/O interface between the microprocessor and external peripheral devices. It provides 24 I/O pins grouped into three 8-bit ports (Port A, Port B, Port C) to facilitate data transfer without requiring dedicated discrete logic hardware.

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4. What are the three modes of operation of the 8255 PPI?

1. **Mode 0 (Basic I/O):** Simple input or output operations without handshaking signals. All ports (A, B, C) can be configured independently as input or output.

2. **Mode 1 (Strobed I/O):** Port A and Port B perform data transfers using dedicated handshake signals from Port C pins to synchronize with peripherals.
3. **Mode 2 (Strobed Bidirectional Bus I/O):** Port A becomes an 8-bit bidirectional bus using 5 pins of Port C for handshaking signals. (Port B can operate in Mode 0 or Mode 1).

5. What is the significance of the Control Word Register in 8255?

The Control Word Register is an internal 8-bit register in the 8255 used to configure its operational modes:

- **Mode Definition Control Word (Bit $D_7 = 1$):** Specifies operating modes (Mode 0, 1, or 2) and data directions (Input/Output) for Ports A, B, and C.
- **BSR Control Word (Bit $D_7 = 0$):** Bit Set/Reset mode, which allows individual setting or clearing of any pin on Port C.

6. How is a Digital-to-Analog (D/A) converter interfaced with an 8086 system?

A D/A converter (such as DAC0808) is connected to an 8086 output port (e.g., 8255 Port A). The 8086 writes binary data to the output port, which passes the digital byte to the DAC input pins. The DAC's internal resistor network (R-2R ladder) converts this binary value into a proportional analog output current or voltage.

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7. What is the role of an Analog-to-Digital (A/D) converter in microprocessor-based systems?

An A/D converter (such as ADC0808) converts physical real-world analog signals (voltage, current, temperature sensor outputs) into digital binary format. The 8086 uses start-of-conversion (SOC) signals to trigger the measurement and reads the digital value when end-of-conversion (EOC) is signaled.

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8. Explain the basic principle of stepper motor interfacing.

A stepper motor converts digital electrical pulses into precise mechanical rotation steps. The microprocessor sends a specific sequence of binary excitation codes (e.g., Wave drive, Half-step, or Full-step sequence) to the motor's stator windings via driver transistors (like ULN2003) to energize the coils in order and rotate the rotor by fixed step angles.

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9. What is the step angle in a stepper motor?

Step angle (θ) is the minimum angular movement achieved by the motor shaft in response to a single input excitation pulse. It is defined as:

$$\text{Step Angle } (\theta) = \frac{360^\circ}{N}$$

(where N is the total number of steps per complete 360° revolution, typically 1.8° or 7.5°).

10. State the main features of the 8254 Programmable Interval Timer (PIT).

- Contains three independent, 16-bit programmable down-counters.
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- Operates from DC to 10 MHz input clock frequencies.
- Supports six software-programmable counter operating modes (e.g., event counter, rate generator, square wave generator).
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- Handles both binary and Binary Coded Decimal (BCD) counting.

11. How many independent counters does the 8254 contain, and what is their size?

The 8254 contains **three independent counters** (Counter 0, Counter 1, and Counter 2). Each counter is **16 bits wide** and operates as a down-counter.

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12. What is the role of the 8259A Programmable Interrupt Controller (PIC)?

The 8259A manages hardware interrupt requests from peripheral devices for the CPU. It handles up to 8 interrupt inputs (IR0–IR7), resolves their relative priorities, masks lower-priority interrupts when requested, and passes the specific 8-bit interrupt vector type to the 8086 during interrupt acknowledge ($\overline{\text{INTA}}$) cycles.

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13. Can 8259A controllers be cascaded? If yes, how many interrupts can be handled?

Yes, 8259A PIC chips can be cascaded in a Master-Slave arrangement. Connecting up to 8 Slave 8259A chips to 1 Master 8259A chip allows the system to expand from 8 interrupt

lines up to a maximum of **64 interrupt inputs**.

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14. What are the key functions of the 8279 Keyboard/Display Controller?

- **Keyboard Interface:** Continuously scans an 8×8 key matrix, handles key debouncing automatically, and stores key codes in an internal 8-byte FIFO RAM.
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- **Display Interface:** Multiplexes and drives up to 16-digit display arrays (7-segment or alphanumeric LEDs) using internal display RAM.
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15. How does the 8279 handle keyboard debouncing?

When a key closure is detected during scanning, the 8279 waits for a debounce period (typically around 10 ms). It verifies if the key remains closed after the delay before generating a key code and writing it to the internal FIFO RAM, ensuring false signals from mechanical bounce are ignored.

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16. What is the purpose of the 8251 USART in microprocessor systems?

The 8251 Universal Synchronous/Asynchronous Receiver Transmitter (USART) facilitates serial data communication. It accepts parallel data from the 8086 data bus and converts it into serial bitstreams for transmission; simultaneously, it accepts incoming serial data and converts it back to parallel data for the CPU.

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17. Define synchronous and asynchronous serial communication supported by 8251.

- **Asynchronous Communication:** Data bits are sent in discrete frames bounded by a Start Bit and one or two Stop Bits, without sharing a common clock line.
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- **Synchronous Communication:** Data blocks are transmitted continuously in sync with a shared clock line, using sync characters at the start of frames to establish frame alignment.
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18. What is the primary function of the 8257 DMA Controller?

The 8257 DMA Controller offloads high-speed data transfers between I/O peripherals (such as disk drives) and system memory. When requested, it takes control of the system address

and data buses from the 8086 CPU to transfer data directly, bypassing CPU instruction execution.

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19. Explain the term "Direct Memory Access" (DMA).

Direct Memory Access (DMA) is a system data transfer mechanism where peripheral hardware transfers data directly to or from main memory without routing the data through CPU registers, maximizing transfer speeds for high-volume I/O.

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20. What are the different channels available in the 8257 DMA controller?

The 8257 DMA controller provides **four independent DMA channels** (Channel 0 through Channel 3). Each channel has dedicated 16-bit Address Registers and 16-bit Terminal Count Registers to direct individual transfer tasks.

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UNIT-4: ADVANCED MICROPROCESSORS (80386DX & 80387)

21. What are the salient features of the 80386DX microprocessor?

- Full 32-bit architecture (32-bit data bus, registers, and ALUs).
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- 32-bit physical address bus enabling access to up to **4 GB** of physical memory.
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- Integrated Memory Management Unit (MMU) supporting segmentation and paging.
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- Supports Virtual Memory (up to 64 Terabytes) and three operating modes.
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22. What is the bus width (data and address bus) of the 80386DX?

- **Data Bus Width:** 32 bits.
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- **Address Bus Width:** 32 bits (enabling a $2^{32} = 4\ \text{GB}$ physical address range).
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23. Name the basic operating modes of the 80386 microprocessor.

1. **Real Address Mode:** Standard startup mode; emulates an 8086 CPU.
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2. **Protected Virtual Address Mode (Protected Mode):** Full 32-bit access with memory protection and paging.
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3. **Virtual 8086 (V86) Mode:** Executed within Protected Mode to run 8086 real-mode programs as isolated tasks.
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24. What is Real Address Mode in 80386?

Real Address Mode is the default operation mode of the 80386 after reset. The processor operates as a fast 8086 processor, limiting physical memory access to 1 MB using 16-bit segment-offset addressing, without hardware protection features.

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25. What is Protected Mode in 80386?

Protected Mode is the native operational mode of the 80386. It provides access to the full 4 GB physical address space, multi-level memory protection (rings 0–3), hardware task switching, virtual memory, and optional paging.

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26. What is Virtual 8086 (V86) Mode?

Virtual 8086 mode is a sub-mode of Protected Mode. It allows the 80386 to execute legacy 16-bit 8086 programs in a protected, multi-tasking environment where the OS can trap and emulate direct hardware accesses.

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27. Explain the concept of segmentation in 80386.

Segmentation divides the linear address space into logical, variable-sized sections called segments (up to 4 GB each). Segment registers hold 16-bit selectors pointing to descriptors in lookup tables that contain the 32-bit base address, limit size, and access privileges.

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28. What is paging in 80386, and what is its page size?

Paging is a hardware address-translation mechanism that maps 32-bit linear addresses into 32-bit physical addresses using fixed-sized memory blocks called **pages**. In the 80386, each page is **4 KB** (4096 bytes).

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29. What is the function of the Descriptor Table in 80386?

A Descriptor Table is a contiguous array of 8-byte segment descriptors stored in memory. The CPU uses these tables to validate memory references, translate logical addresses, and enforce security checks in Protected Mode.

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30. Define Global Descriptor Table (GDT) and Local Descriptor Table (LDT).

- **Global Descriptor Table (GDT):** A master system table containing segment descriptors accessible by all software tasks in the system.
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- **Local Descriptor Table (LDT):** A task-specific table containing descriptors private to an individual application task for memory isolation.
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31. What are the register categories in the 80386 register organization?

1. **General-Purpose Registers:** Eight 32-bit registers (EAX, EBX, ECX, EDX, ESI, EDI, EBP, ESP).
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2. **Segment Registers:** Six 16-bit selectors (CS, DS, SS, ES, FS, GS).
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3. **Control Registers:** CR0, CR1, CR2, CR3 (manage flags, paging, and protection).
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4. **System Address Registers:** GDTR, LDTR, IDTR, TR.
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5. **Instruction Pointer & Flags:** EIP and EFLAGS.
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32. What data types are supported by the 80386 architecture?

- **Standard Integer Data Types:** Bytes (8-bit), Words (16-bit), Doublewords (32-bit).
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- **Bit Data Types:** Individual bits, Bit Strings (up to 4 GB).
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- **String Data Types:** Byte, Word, and Doubleword strings.
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- **BCD Data Types:** Packed and Unpacked BCD.
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- **Pointer Data Types:** 48-bit Far Pointers (16-bit selector + 32-bit offset).
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33. What are the addressing modes available in 80386?

- **Register & Immediate**
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- **Direct Address**
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- **Register Indirect**
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- **Based Addressing**
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- **Indexed Addressing**
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- **Scaled Indexed Addressing:** $\text{Effective Address} = \text{Base} + (\text{Index} \times \text{Scale Factor}) + \text{Displacement}$ *(where Scale Factor = 1, 2, 4, or 8).*
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34. What is the purpose of the Task State Segment (TSS) in 80386?

A Task State Segment (TSS) is a dedicated memory structure that stores a task's hardware execution state (register values, stack pointers for privilege levels 0–2, and LDT selector). It enables automatic hardware-level task switching.

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35. What is privilege level (Ring 0 to Ring 3) in 80386 protected mode?

The 80386 implements a 4-ring hierarchical protection model to isolate software components:

- **Ring 0 (Highest):** OS Kernel / Core drivers (full system access).
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- **Ring 1:** OS Services / System extensions.
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- **Ring 2:** System Utilities / Custom drivers.
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- **Ring 3 (Lowest):** User Application programs (restricted access).
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36. What is the role of the 80387 numeric coprocessor?

The 80387 is a dedicated math coprocessor designed to handle floating-point arithmetic and high-precision calculations (IEEE 754 floating-point standard). It operates concurrently with the 80386 CPU to offload complex mathematical tasks.

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37. How does the 80387 interface with the 80386 processor?

The 80387 connects to the 80386 via the shared system data bus and dedicated handshake lines:

- $\mathrm{BUSY\#}$ / $\mathrm{ERROR\#}$: Signals processing status and execution errors.
- PEREQ (Processor Extension Request): Requests the 80386 CPU to transfer coprocessor operands to or from memory.
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38. What is the size of the internal data registers of the 80387 coprocessor?

The 80387 features an internal register stack consisting of **eight 80-bit wide floating-point registers** ($ST(0)$ through $ST(7)$), supporting extended precision calculations.

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39. Mention any two differences between 8086 and 80386 architecture.

1. **Bus Width:** The 8086 has a 16-bit data bus and a 20-bit address bus (1 MB physical memory), while the 80386DX has a 32-bit data bus and a 32-bit address bus (4 GB physical memory).
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2. **Operating Modes & MMU:** The 8086 operates only in real mode, whereas the 80386 integrates an MMU supporting Real Mode, Protected Mode, Paging, and Virtual 8086 Mode.
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40. What is the purpose of control registers (CR0 to CR3) in 80386?

- **CR0:** Contains system control flags, including PE (Protection Enable) and PG (Paging Enable).
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- **CR1:** Reserved for future intel architecture extensions.
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- **CR2:** Page Fault Linear Address Register (stores the 32-bit address that caused a page fault).
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- **CR3:** Page Directory Base Register (PDBR) (holds the physical base address of the Page Directory).
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- Would you like detailed block diagrams or pin descriptions for any of the peripheral ICs (8255, 8254, 8259A, 8257, 8251)?
- Do you need long-answer (10-mark / 15-mark) analytical questions and answers for Unit-3 or Unit-4?

- Are there specific topics within Unit-3 or Unit-4 where you would like additional step-by-step assembly program examples (e.g., 8255 control word initialization or 8254 timer configuration)?